

ACHINTYA RAI

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EDUCATION

University of California, San Diego

Bachelor of Science in Computer Engineering

Sept. 2024 – June 2028

GPA: 3.9/4.0

- Coursework: Advanced Data Structures & Algorithms, Machine Learning, Linear Systems, Computer Architecture, Discrete Mathematics

WORK EXPERIENCE

Adobe

Software Engineering Intern

San Jose, CA

June 2026 – Sept. 2026

- Profiled **60.6M+** payment events in Databricks, identifying **86,387** duplicate records and a **47%** null rate in timestamp fields to establish the correct data source for downstream monitoring.
- Identified a **26-hour** processing outage in Feb 2026 via hourly volume analysis across **1,272** payment monitoring segments, establishing it as the primary validation benchmark for the anomaly detector.
- Designed a time-series anomaly detector, iterating over statistical algorithms to select a machine learning model handling multi-layer seasonality across **10+** workflows and **30+** countries.
- Developed SQL pipelines and an MLOps pipeline to schedule weekly retraining and hourly inference across all machine learning forecasting models.

VALT Health

Founding Software Engineer

Remote

Aug. 2025 – Feb. 2026

- Engineered a Flask backend to handle **100k+** datapoints from fitness devices (Fitbit, Apple Watch, etc). Designed database tables and schemas to unify data formats across 20+ distinct sources.
- Implemented asynchronous batch processing and storage workflows in GCP, yielding a **35%** increase in execution speed and **60%** reduction in redundant database write operations.
- Automated local environment setup and data seeding via Bash scripts, reducing testing time by up to **80%**.
- Authored technical specifications and documentation standards to reduce on-boarding friction for new engineers.

Decklar (formerly Roambee)

Software Engineering Intern

Sunnyvale, CA

June 2024 – Aug. 2024

- Developed real-time shipment monitoring modules for **Amazon, Coca-Cola, and Samsung**, optimizing Python and NumPy data processing pipelines to achieve a **15%** capacity boost for **1M+** weekly events.
- Implemented advanced data structures for visualization and analytics modules, resulting in **20%** faster reporting speeds and improved data integrity across internal shipment dashboards.
- Enhanced workflow efficiency by **15%** through the Panels-based UI/UX refinements and iterative user testing.

RESEARCH

Graph Theory & Neural Architecture

Undergraduate Research Fellow (NSF REU)

San Diego, CA

March 2026 – Present

- Engineered a Python pipeline to analyze LLM embedding manifolds (Llama-2, DeepSeek-V2), implementing weight-matrix extraction utilities to reduce data footprints from **500GB** checkpoints to **13GB** localized manifolds.
- Optimized architectural placement scripts and normalized preprocessing, achieving a **30x runtime reduction** (30 mins to <1 min) while enabling large-scale manifold analysis on resource-constrained **8GB RAM** systems.
- Utilized **Rent's Rule** and Louvain Clustering to identify model "backbones," discovering that noise-filtering increases Inter-Cluster Network (ICN) density from **0.18%** to **13.25%** in Llama-2-70B.

Data Science Teaching & Learning Lab

Undergraduate Researcher

San Diego, CA

Sept. 2024 – Present

ChatSight | *Python, FastAPI, Gemini API*

- Developed a data-wrangling tool for instructors to analyze student AI usage, implementing an LLM-augmented workflow with the Gemini API to automate the labeling of **16k+** data points.

- Reduced labeling time by **90%** through three iterations of user-centric UI/UX refinements, optimizing the interface for ease of use and high-throughput human-in-the-loop verification.

WISE Jupyter AI Tutor | *Python, Data Validation, JupyterLab, OpenAI API*

- Architected a multi-agent AI tutoring framework using Python to provide real-time academic support, implementing hallucination-reduction strategies to ensure 95%+ accuracy in response generation.
- Developed high-throughput agentic workflows to rapidly process high-density Jupyter Notebooks, synthesizing **100+** computational cells and datasets of **100k+** **points** to extract real-time insights.

Applications of Multi-Agent Systems

Undergraduate Research Fellow (UC Scholars)

San Diego, CA

June 2025 – Aug. 2025

Multi-agent AI Chess Tutor | *Gemini API, Django, React, SQLite*

- Built AI Chess Analysis Tool using **Django (Python)**, **React (TypeScript)**, and **SQLite** via Agile sprints.
- Architected **Multi-Agent workflows** reducing hallucinations by **80%** with an **83% faster** system runtime.
- Developed **7 REST API endpoints** and production prototypes validated by **15+** chess players and coaches.
- Published and presented AI research at the **UCSD Summer Research Conference (SRP)**.

PROJECTS

Watchdog | *Python, Gemini API, Browser-Use, Asyncio, SQLite* 

April 2026

- Engineered an agentic browser safeguard using the **browser-use SDK** and **Gemini Vision** to autonomously monitor tabs and intervene on unproductive sites via GUI automation.
- Developed a concurrent CLI with **Asyncio** to manage real-time URL blacklisting and **Notion API** task synchronization, reducing context-switching friction.
- Optimized local state persistence using **aiosqlite** for non-blocking database operations, managing relational mappings between browser profiles and active task groups.

NanoGPT | *Python, PyTorch, C++, Transformers, LLM Optimization* | 

Mar. 2026

- Implemented a GPT-style transformer from scratch in **Python** and **PyTorch**, implementing multi-headed attention, causal masking, and backpropagation layers.
- Optimized model memory layout and compiled critical sub-modules in **C++**, reducing per-token latency by **25%** on resource-constrained systems.
- Engineered scalable dataloaders and pre-processing scripts to stream tokenized datasets, maintaining **100%** hardware utilization during training runs.

SquaredUpChess | *React, TypeScript, Django, Websockets, AWS, Coolify, Docker* 

Aug. 2025 – Dec. 2025

- Engineered a full-stack platform using **AWS** to facilitate coach-student scheduling, real-time lesson tracking, and payment processing.
- Developed a responsive UI with **React**, implementing state management for low-latency, real-time updates.
- Architected **20+ Django REST API endpoints** to handle secure authentication, session management, and transaction logic.
- Optimized **SQLite** performance through strategic indexing and query tuning to maintain multi-user scalability.
- Established a **CI/CD pipeline** with comprehensive error logging and automated testing to ensure 99.9% deployment reliability.

MovieRecapinator | *Java, Spring Boot, Maven, JavaScript, IMDb API* 

April 2025

- Developed a recommendation engine using **Spring Boot** to process **700,000+** **entries** from the IMDb API, identifying high-priority episodes for series recaps.
- Collaborated with a 4-person team at **DataHacks** to architect the full-stack system, delivering a functional MVP with Java and JavaScript within a 24-hour sprint.

TECHNICAL SKILLS

Languages: Python, C/C++, TypeScript, SQL (Postgres/SQLite), Java, JavaScript, Bash, HTML/CSS

Development: React, Django, Node.js, FastAPI, Asyncio, WebSockets, Flask, JUnit, PyTest

AI & Cloud: PyTorch, AWS, GCP, Gemini/OpenAI API, Docker, GitHub Actions, browser-use, Linux